MILITARY SPECIFICATION

TEST KIT, MK-91(*)/APM-66

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be Test Kit MK-91(*)/APM-66, (part of Test Set, Radar, AN/APM-66), to be used to facilitate the test, maintenance, and calibration of major units of Radar Set, AN/APN-22.

(*) The parenthesis in the type designation shall be replaced by a capital letter by the procuring agency upon application by the contractor in accordance with the requirements of paragraph 3.2.11 after each contract has been awarded. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

1.2 Classification: The MK-91(*)/APM-66 shall be of one type and shall consist of:

Items	Type Designation	Max. Allow. Size (Inches) L	H	D	Max. Allow. Wt. (Lbs.)	Appl. Para.
Cable Assembly, Special Purpose, Electrical	CX-2088/APM-66	49			4	3.3.1
Cable Assembly, Special Purpose, Electrical	CX-2089/APM-66	73-1/2			5	3.3.2
Test Stand	MT-1138/APM-66	25	6	6	5	3.3.3
Test Stand	MT-1139/APM-66	6	6-1/2	6	1	3.3.4
Case, Test Kit	CY-1207/APM-66	27	12	12	22	3.3.5

1.3 Accessories - The MK-91(*)/APM-66 may be used with the following equipment which are not required to be furnished with the test kit:

Test Set, Altimeter; TS-745/APM-66
Calibrator, Radar Altimeter; TS-746/APM-66
Multimeter, ME-6D/U or equivalent
Voltmeter, TS-505/U or equivalent
Oscilloscope, AN/USM-24 or equivalent
Multimeter, AN/PSM-4 or equivalent

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of invitation for bids, shall form a part of this specification:

SPECIFICATIONS

Military

MIL-T-21200	Test Equipment, For Use with Electronic and Fire Control Systems; General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits and Tests, Aircraft Electrical and Electronic Equipment
MIL-H-15362	Harnesses; Test Bench, Electronic; General Specification for
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

Bureau of Aeronautics

SAR-300	Material Changes and Material Bulletins; Preparation of

STANDARDS

Military

MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

Air Force-Navy Aeronautical Standard

AN-3108	Connectors; Electrical, Plugs, Angle 90°

2.2 Availability of Documents:

(1) When requesting applicable documents, refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania, (Attention: Code CDS).

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2.3 Precedence - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over all referenced documents.

(2) This Specification - This specification shall have precedence over all subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - The equipment shall fulfill the following requirements for parts and materials:

3.1.1 Selection of Parts and Materials - The fulfillment of major design objectives shall govern the selection of parts and materials. All parts and materials shall meet the requirements of Specification MIL-T-21200; however, if the contractor can demonstrate the use of such parts and materials will not fulfill the major design objectives, then parts and materials shall be used which most nearly meet or exceed the requirements of Specification MIL-T-21200.

3.1.2 Approval for Nonstandard Parts and Materials - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200. Requests for approval must be forwarded to the procuring agency by the contractor before parts are purchased or fabricated.

3.2 General Requirements

3.2.1 General - The equipment shall conform with all applicable design, construction and workmenship requirements of Specification MIL-T-21200 unless otherwise specified.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions, or reasonable combination of service conditions, as specified in Specification MIL-H-15362 except that the requirements specified shall apply only within the temperature range from -40° to +55° C.

3.2.3 Total Weight - The total weight of the equipment shall not be greater than 50 pounds.

3.2.4 Power Requirements - The equipment shall be designed to handle all voltages and currents necessary for the test, maintenance, and calibration of the AN/APN-22.

3.2.5 Operating Life

3.2.5.1 Reliable Operating Life - The equipment shall be capable of operation, as specified herein, for a period of 200 hours without requiring servicing or replacement of parts. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

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3.2.5.2 Total Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. The life of such parts and parts requiring replacement within this interval shall be specified by the manufacturer.

3.2.6 Operating Stability - The equipment shall operate with optimum performance for at least 200 hours, continuously or intermittently.

3.2.7 Interference - The equipment shall contain a sufficient number of shielded, single, double, and triple conductors to minimize radiated or conducted interference to or from the signal conductors in accordance with the requirements of Specification MIL-I-6181. Reasonable precautions shall be taken to prevent the unshielded wires from producing fields of sufficient intensity to interfere with the satisfactory performance of other normally installed electronic equipment.

3.2.8 Cables and Connectors - Cables and connectors shall be in accordance with the requirements of Specification MIL-T-21200 except that special connectors for intra-unit and interunit connectors may be used providing the design and workmanship conform to good engineering practices, and the performance of the equipment is not affected.

3.2.9 Fungus and Moisture Treatment - The equipment shall be moisture and fungus-proofed by selection of parts and materials nonnutrient for fungus. The parts and materials shall be treated so that prior to their use in the equipment, over-all spraying of the equipment is not necessary.

3.2.10 Interchangeability - Mechanical and electrical interchangeability shall exist between similar assemblies, subassemblies, and replaceable parts regardless of the manufacturer or supplier. Interchangeability for the purposes of this paragraph does not mean identity, but requires that a substitution of such like assemblies, subassemblies, and replaceable parts be easily effected without physical or electrical modifications to any part of the equipment including cabling, wiring, and mounting. In the design of the equipment, provision shall be made for design tolerance to accommodate various components having limiting dimensions and characteristics set forth in the specification for the particular components involved without departure from the specified performance.

3.2.11 Nameplates and Nomenclature - Nameplates and nomenclature shall be in accordance with the requirements of Specification MIL-N-18307.

3.3 Detail Requirements

3.3.1 Cable Assembly, Special Purpose, Electrical CX-2088/APM-66 - Two CX-2088/APM-66 unit test cable assemblies shall be provided as a convenient means for testing individual units of Electronic Control Amplifier AM-291/APN-22, when it is desired to remove such a unit from the main chassis.

3.3.1.1 Form Factor - The over-all length of each cable assembly shall not be greater than 49 inches. The weight shall not be greater than 4 pounds. The form factor shall permit placement of units under test not less than 24 inches apart on a flat surface.

3.3.1.2 Contents - Each assembly shall contain a 19-conductor cable, two special 20-pin receptacles and one 20-pin male connector.

3.3.2 Cable Assembly, Special Purpose, Electrical CX-2089/APM-66 - One CX-2089/APM-66 interconnecting cable assembly shall be provided for connecting power sources to Electronic Control Amplifier AM-291/APN-22, and power, signal, and control voltages from this unit to Height Indicator ID-257/APN-22 and Receiver-Transmitter RT-160/APN-22.

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3.3.2.1 Form Factor - The cable assembly shall consist of three cables terminated in a junction box. The length of the assembly shall not be greater than 49 inches except that the cable connection to the RT-160/APN-22 shall be 73-1/2 inches long maximum. The weight of the assembly shall not be greater than 5 pounds. The form factor shall permit placement of units under test not less than 36 inches apart on a flat surface.

3.3.2.2 Junction Box - The junction box shall terminate the three cables, and shall contain two 20-pin female and two 20-pin male connectors, a 3-ampere fuse in the 28-V d-c line, a 0.3 ampere fuse in the 115-V a-c regulated line, and a 1.5-ampere fuse in the 115-V a-c unregulated line.

3.3.2.3 Cables - The 6-conductor power cable shall terminate at stripped wire ends. The 8-conductor indicator cable shall terminate in an AN3108B-18 receptacle. The 12-conductor antenna cable shall terminate in a special 15-pin female receptacle capable of mating with the P-2001 connector on the Radar Receiver-Transmitter RT-160/APN-22.

3.3.2.3.1 Power Cable Wire Workings - The stripped wire ends of the 6-conductor power cable shall be color coded as follows:

+28V, DC	Red
D-C Ground	Black
115 V, AC (380 to 420 yds)	Yellow
A-C Ground (380 to 420 yds)	Orange
115 V, AC (320 to 1000 yds)	Green
A-C Ground (320 t0 1000 yds)	Blue

3.3.3 Test Stand MT-1138/APM-66 - One MT-1138/APM-66 amplifier test stand shall be provided as a suitable means for mounting Electronic Control Amplifier AM-291/APN-22 for test purposes.

3.3.3.1 Form Factor - The test stand shall contain a base, side channel assembly, front stay, continuous hinge, and associated hardware. The base shall be hinged along one side to the side channel assembly (on which the amplifier is mounted) to permit rotation of the equipment 90 degrees along its longitudinal axis. The over-all dimensions (side channel latched to base) shall not be greater than 25 inches long, 6 inches wide, and 6 inches deep. The weight shall not be greater than 5 pounds.

3.3.4 Test Stand MT-1139/APM-66 - One MT-1139/APM-66 indicator test stand shall be provided for securing the Height Indicator ID-257/APN-22 to a test bench.

3.3.4.1 Form Factor - The bracket shall be L-shaped, bent at 90 degrees, and shall have adequate holes for mounting the indicator. The over-all dimensions shall not be greater than 6 inches long, 6 inches wide, and 6-1/2 inches high. The weight shall not be greater than 1 pound.

3.3.5 Case, Test Kit CY-1207/APM-66 - One CY-1207/APM-66 shall be provided in accordance with the requirements of Specification MIL-T-21200.

3.3.5.1 Form Factor - The case shall be of rectangular construction. The over-all dimensions shall not be greater than 27 inches long, 12 inches wide, and 12 inches high. The weight shall not be greater than 22 pounds.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.4. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4.2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-21200. The design and test data shall include the data obtained by the contractor in testing to Specifications MIL-T-5422 and MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's complaince with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 except the vibration tests which shall conform to MIL-STD-167 (Type I). Interference tests shall be in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

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(2) Vibration Test (90 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.5 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.6 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Bureau of Aeronautics.

4.7 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-300.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for shipment in accordance with the requirements of Specification MIL-P-17555.

6. NOTES

6.1 Ordering Information - Invitation for bids and contracts or orders shall specify the title, number, and date of this specification.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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